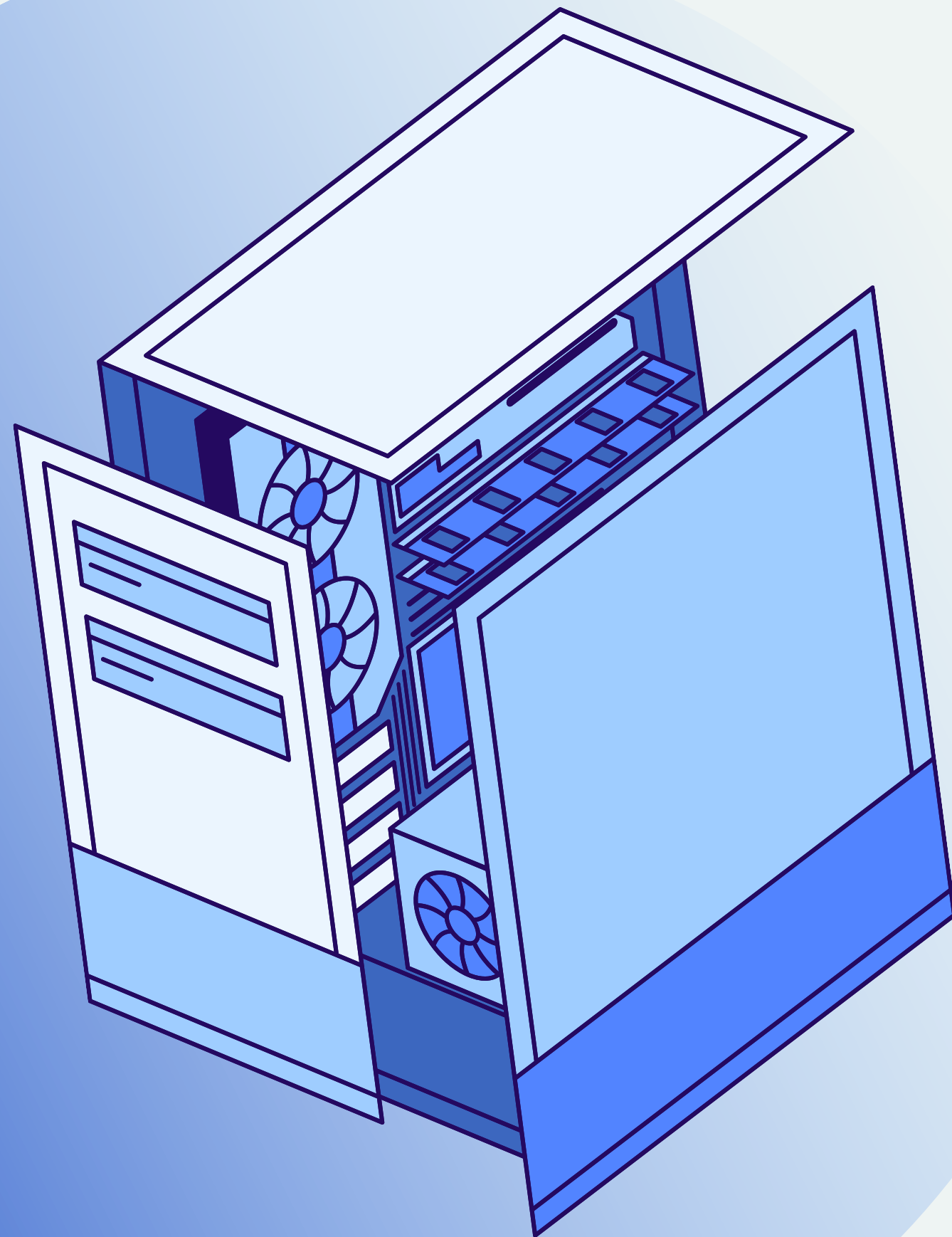




MACHINE LEARNING 1

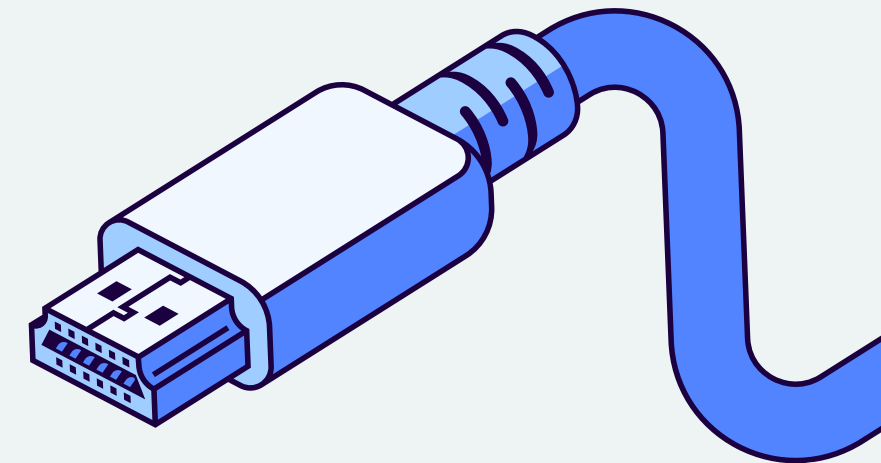
TITANIC COMPETITION

PRESENTATION

Bastian ARDILLON

Octave SAVEAUX

Martin BRULEY



PROBLEM & DATASET

- Goal: Predict passenger survival on the Titanic
- Kaggle dataset:
 - Training set: 891 passengers
 - Test set: 418 passengers
- Target variable: Survived (binary classification)



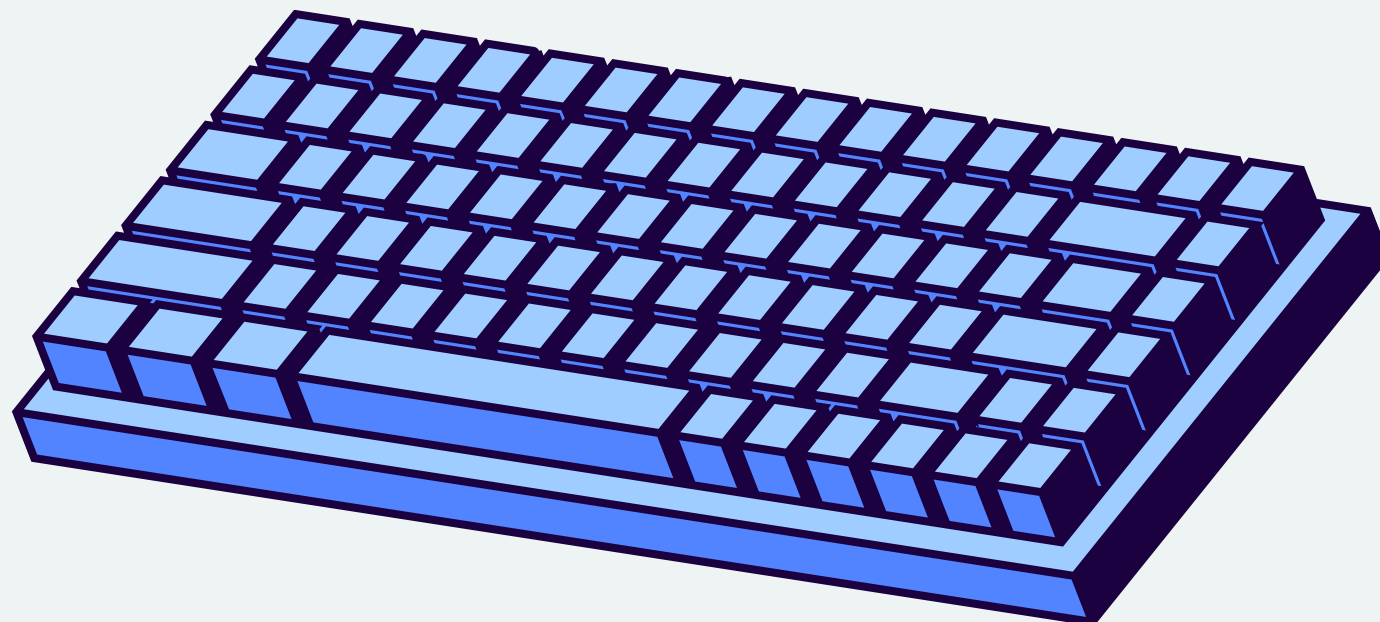
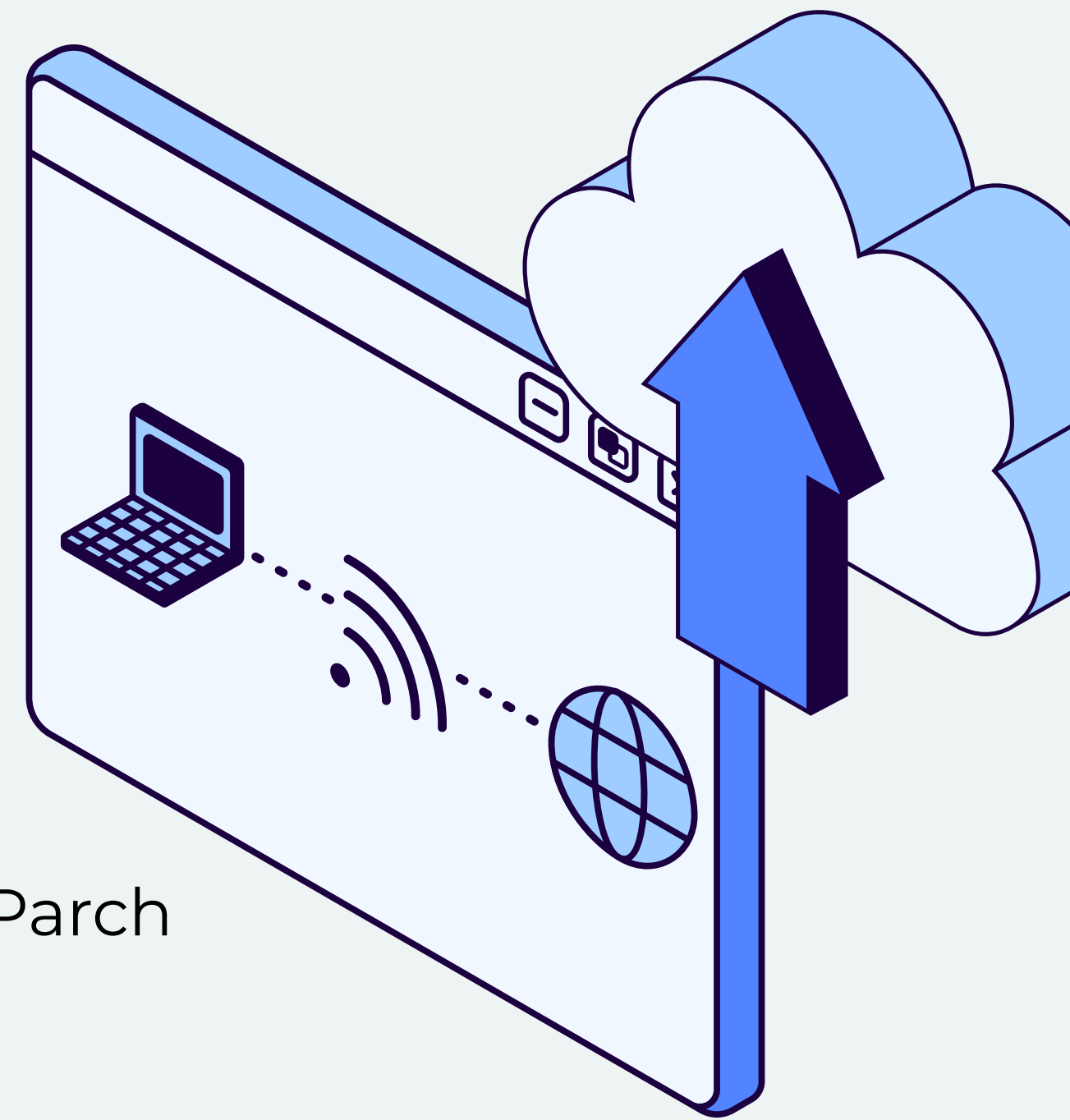
DATASET MERGING STRATEGY



- Train and test datasets were merged into a single dataset
- Purpose:
 - Apply identical preprocessing
 - Avoid data leakage or feature mismatch
- After preprocessing:
 - Dataset split back into train and test sets

HANDLING MISSING VALUES

- Age: filled with median value
- Embarked: filled with most frequent value
- Fare: filled with median grouped by Pclass, SibSp and Parch
- Cabin: dropped due to too many missing values



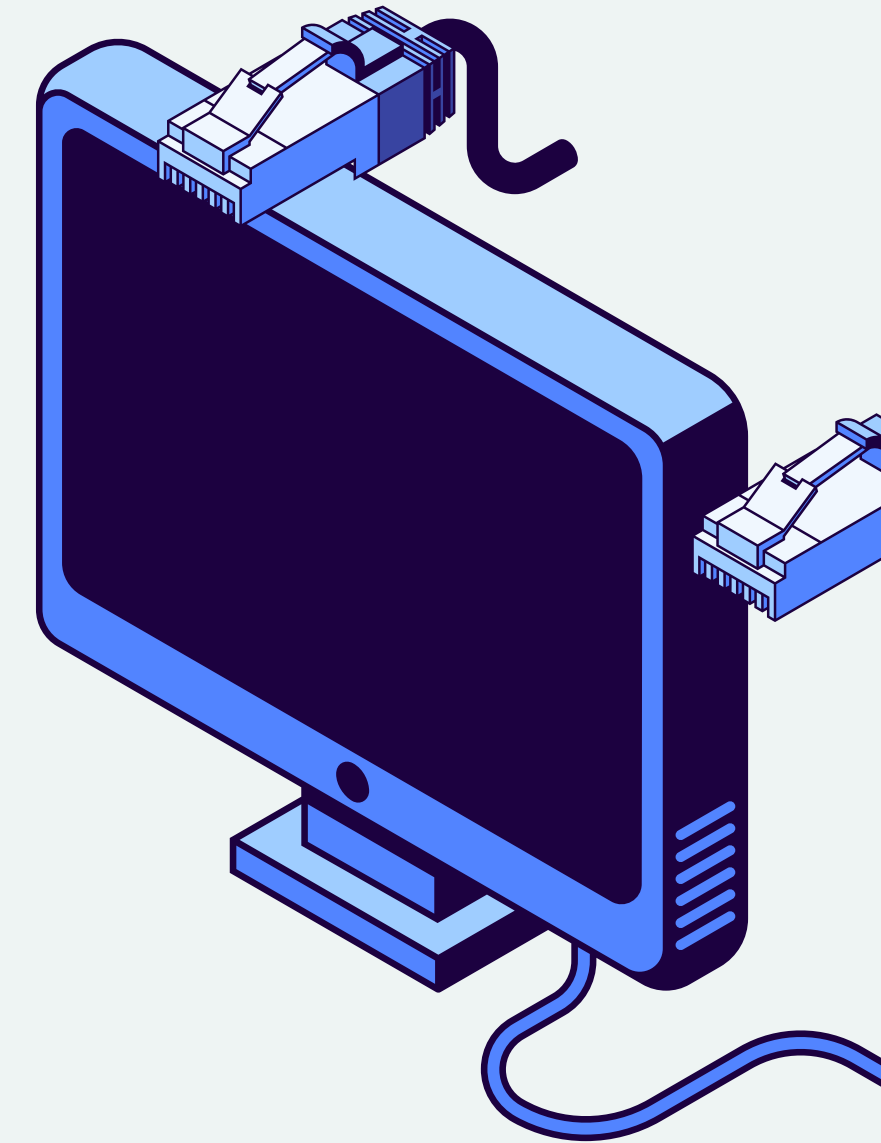
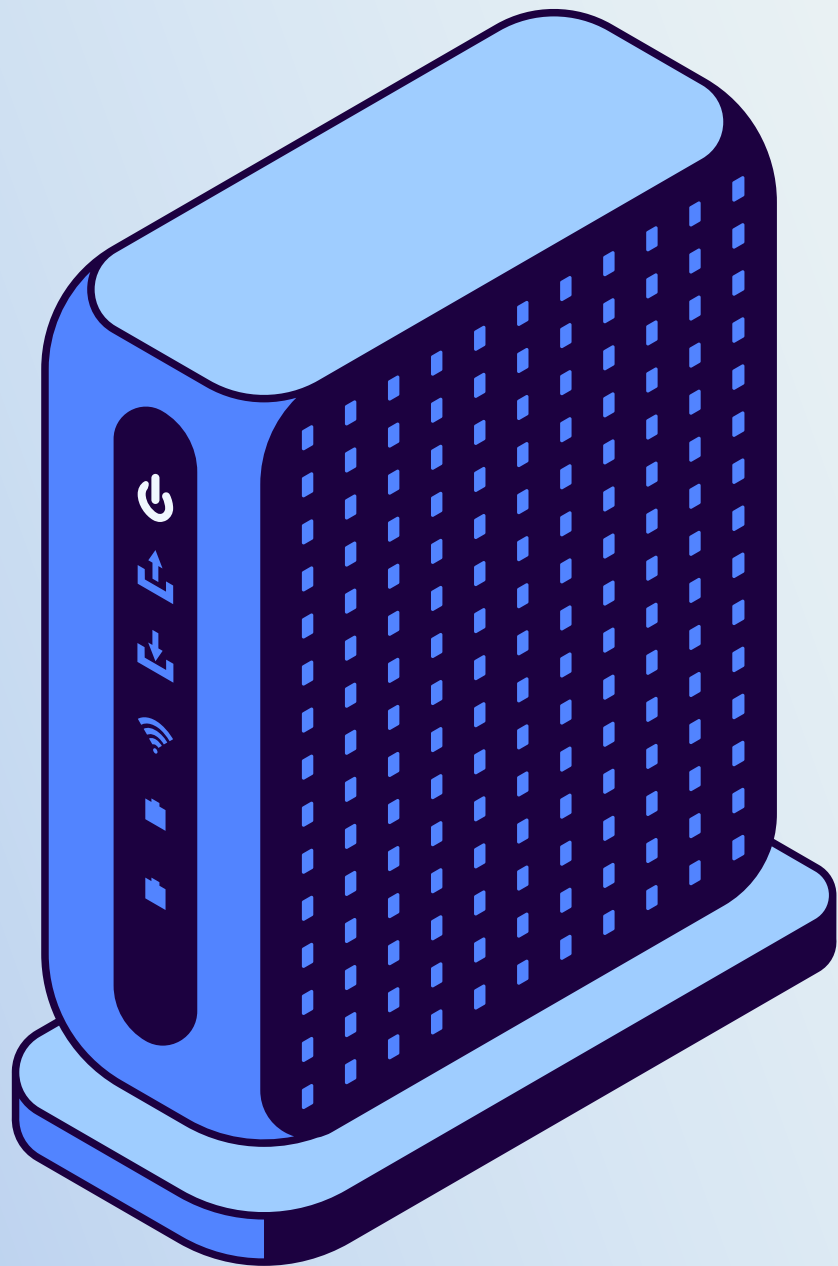
FEATURE ENGINEERING & ENCODING / SCALING

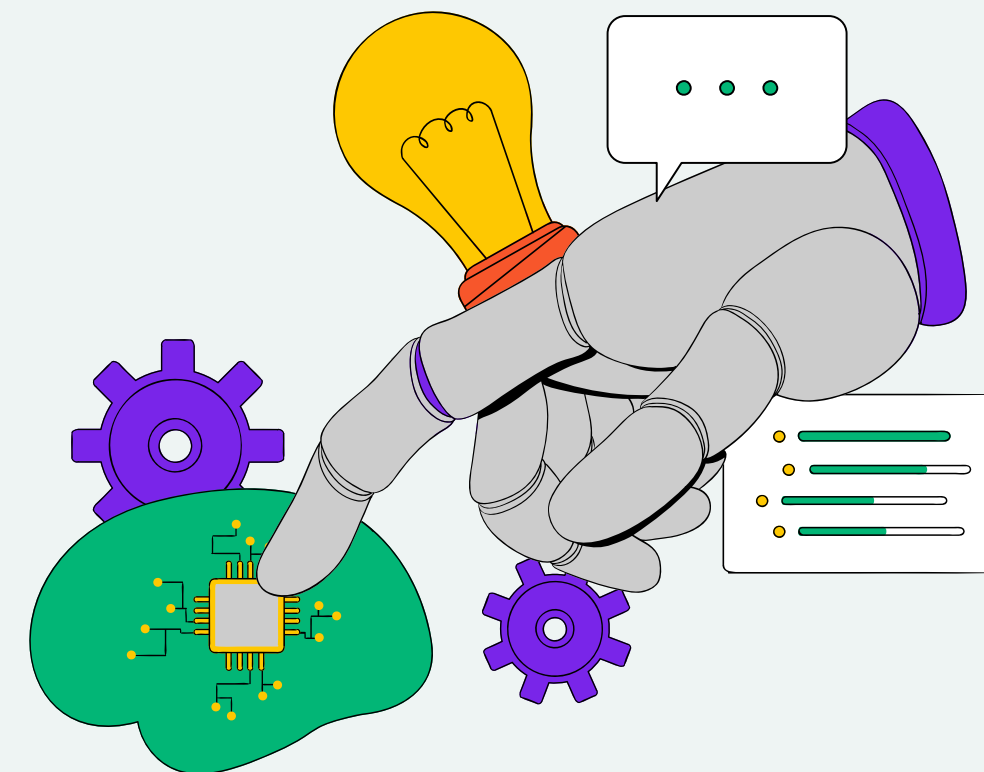
- $\text{FamilySize} = \text{SibSp} + \text{Parch} + 1$
- IsAlone: binary indicator
- Title: extracted from passenger names
- Rare titles grouped into one category
- Sex: binary encoding
- Embarked: numerical encoding
- One-hot encoding for categorical features
- Numerical features scaled using StandardScaler



MODELS TESTED & RESULTS

- Logistic Regression (baseline)
- Gradient Boosting Classifier
- Random Forest Classifier
- Best validation accuracy: ~78,708%





THANK YOU